

3 VRFs and BGP-MPLS L3VPNs

3.1 Network scenario and configuration

The topology replicates the Lab 2.2 scenario: two organizations, *blue* and *red*, with two sites each, connected through a BGP/MPLS L3VPN provider.

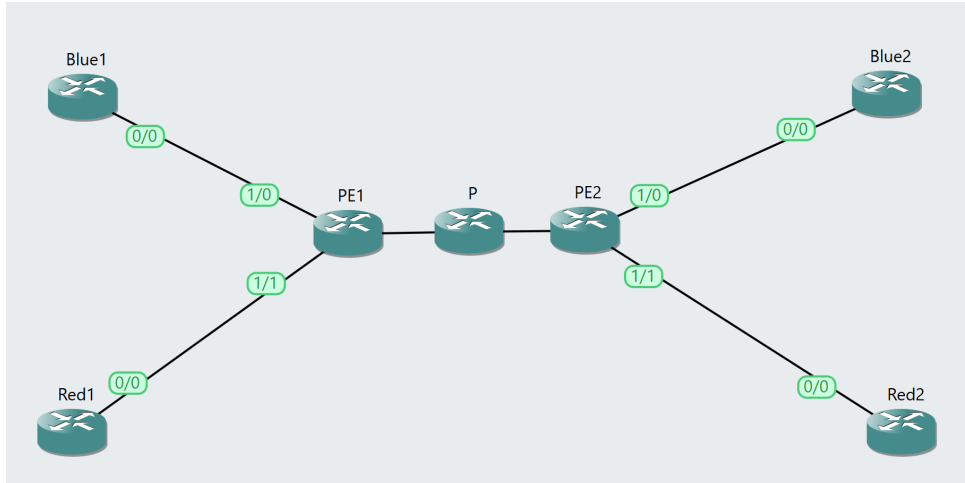


Figure 28: Logical topology built in GNS3.

PE1 and PE2 were configured as Cisco 7200 routers; P, Blue1, Blue2, Red1, and Red2 as Cisco 3725 routers [2]. The core uses OSPF, LDP/MPLS, and MP-BGP VPNv4.

Table 4: Addressing plan used in the simulation.

Router	Interface	Address	Role
Blue1	Fa0/0	10.10.1.100/24	CE blue, site 1
Blue1	Lo0	192.168.101.1/24	Simulated blue network, site 1
Blue2	Fa0/0	10.10.2.100/24	CE blue, site 2
Blue2	Lo0	192.168.102.1/24	Simulated blue network, site 2
Red1	Fa0/0	11.11.1.100/24	CE red, site 1
Red1	Lo0	192.168.111.1/24	Simulated red network, site 1
Red2	Fa0/0	11.11.2.100/24	CE red, site 2
Red2	Lo0	192.168.112.1/24	Simulated red network, site 2
PE1	Fa1/0	10.10.1.1/24	VRF blue
PE1	Fa1/1	11.11.1.1/24	VRF red
PE1	Fa0/0	222.222.1.1/24	Provider core
PE1	Lo0	1.1.1.1/32	Router ID MP-BGP/LDP
P	Fa0/0	222.222.1.2/24	Provider core
P	Fa0/1	222.222.2.2/24	Provider core
PE2	Fa0/0	222.222.2.10/24	Provider core
PE2	Fa1/0	10.10.2.1/24	VRF blue
PE2	Fa1/1	11.11.2.1/24	VRF red
PE2	Lo0	2.2.2.2/32	Router ID MP-BGP/LDP

Each CE has one interface toward the PE, one Loopback that represents the customer network, and a default route to the respective PE.

```
Blue1
interface FastEthernet0/0
 ip address 10.10.1.100 255.255.255.0
 no shutdown
interface Loopback0
 ip address 192.168.101.1 255.255.255.0
 ip route 0.0.0.0 0.0.0.0 10.10.1.1
```

Listing 33: CE router configuration on Blue1.

```
Blue2
interface FastEthernet0/0
  ip address 10.10.2.100 255.255.255.0
  no shutdown
interface Loopback0
  ip address 192.168.102.1 255.255.255.0
ip route 0.0.0.0 0.0.0.0 10.10.2.1
```

Listing 34: CE router configuration on Blue2.

```
Red1
interface FastEthernet0/0
  ip address 11.11.1.100 255.255.255.0
  no shutdown
interface Loopback0
  ip address 192.168.111.1 255.255.255.0
ip route 0.0.0.0 0.0.0.0 11.11.1.1
```

Listing 35: CE router configuration on Red1.

```
Red2
interface FastEthernet0/0
  ip address 11.11.2.100 255.255.255.0
  no shutdown
interface Loopback0
  ip address 192.168.112.1 255.255.255.0
ip route 0.0.0.0 0.0.0.0 11.11.2.1
```

Listing 36: CE router configuration on Red2.

In the provider core, P only transports IP/MPLS. Its global table contains the core links and the PE Loopbacks learned through OSPF.

```
P#show running-config | subsection router ospf|interface FastEthernet
interface FastEthernet0/0
  ip address 222.222.1.2 255.255.255.0
  mpls ip
  no shutdown
interface FastEthernet0/1
  ip address 222.222.2.2 255.255.255.0
  mpls ip
  no shutdown
router ospf 1
  network 222.222.1.0 0.0.0.255 area 0
  network 222.222.2.0 0.0.0.255 area 0

P#show ip route
  1.0.0.0/32 is subnetted, 1 subnets
O    1.1.1.1 [110/11] via 222.222.1.1, FastEthernet0/0
C    222.222.2.0/24 is directly connected, FastEthernet0/1
  2.0.0.0/32 is subnetted, 1 subnets
O    2.2.2.2 [110/11] via 222.222.2.10, FastEthernet0/1
C    222.222.1.0/24 is directly connected, FastEthernet0/0

P#show ip ospf neighbor
Neighbor ID    Pri   State           Address         Interface
2.2.2.2        1     FULL/BDR        222.222.2.10   FastEthernet0/1
```

```

1.1.1.1          1    FULL/BDR          222.222.1.1      FastEthernet0/0

P#show mpls ldp neighbor
Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 222.222.1.2:0
  State: Oper; Msgs sent/rcvd: 10/10; Downstream
Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 222.222.1.2:0
  State: Oper; Msgs sent/rcvd: 10/10; Downstream

```

Listing 37: Router P configuration and OSPF/LDP validation.

On PE1 and PE2, two VRFs were created. The Route Distinguisher distinguishes the VPNv4 routes; the Route Target controls import/export. MP-BGP uses the VPNv4 address family and sends extended communities [4].

```

PE1
ip vrf blue
  rd 1:1
  route-target both 1:1
ip vrf red
  rd 2:2
  route-target both 2:2
mpls ldp router-id Loopback0 force
interface FastEthernet1/0
  ip vrf forwarding blue
  ip address 10.10.1.1 255.255.255.0
  no shutdown
interface FastEthernet1/1
  ip vrf forwarding red
  ip address 11.11.1.1 255.255.255.0
  no shutdown
interface FastEthernet0/0
  ip address 222.222.1.1 255.255.255.0
  mpls ip
  no shutdown
interface Loopback0
  ip address 1.1.1.1 255.255.255.255
router ospf 1
  network 222.222.1.0 0.0.0.255 area 0
  network 1.1.1.1 0.0.0.0 area 0
ip route vrf blue 192.168.101.0 255.255.255.0 10.10.1.100
ip route vrf red 192.168.111.0 255.255.255.0 11.11.1.100
router bgp 100
  neighbor 2.2.2.2 remote-as 100
  neighbor 2.2.2.2 update-source Loopback0
  address-family vpnv4
    neighbor 2.2.2.2 activate
    neighbor 2.2.2.2 send-community extended
  address-family ipv4 vrf blue
    redistribute static
  address-family ipv4 vrf red
    redistribute static

```

Listing 38: VRF and MP-BGP configuration on PE1.

```

PE2
ip vrf blue
  rd 1:1
  route-target both 1:1
ip vrf red
  rd 2:2
  route-target both 2:2
mpls ldp router-id Loopback0 force

```

```

interface FastEthernet1/0
 ip vrf forwarding blue
 ip address 10.10.2.1 255.255.255.0
 no shutdown
interface FastEthernet1/1
 ip vrf forwarding red
 ip address 11.11.2.1 255.255.255.0
 no shutdown
interface FastEthernet0/0
 ip address 222.222.2.10 255.255.255.0
 mpls ip
 no shutdown
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
router ospf 1
 network 222.222.2.0 0.0.0.255 area 0
 network 2.2.2.2 0.0.0.0 area 0
ip route vrf blue 192.168.102.0 255.255.255.0 10.10.2.100
ip route vrf red 192.168.112.0 255.255.255.0 11.11.2.100
router bgp 100
 neighbor 1.1.1.1 remote-as 100
 neighbor 1.1.1.1 update-source Loopback0
 address-family vpnv4
  neighbor 1.1.1.1 activate
  neighbor 1.1.1.1 send-community extended
 address-family ipv4 vrf blue
  redistribute static
 address-family ipv4 vrf red
  redistribute static

```

Listing 39: VRF and MP-BGP configuration on PE2.

The iBGP session between the PE Loopbacks was established. The `State/PfxRcd` field is 0 in the global IPv4 unicast table because the lab routes are transported in the VPNv4 address family.

```

PE1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 100
BGP table version is 1, main routing table version 1

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
2.2.2.2	4	100	7	8	1	0	0	00:00:51	0

Listing 40: BGP session between PE1 and PE2.

3.2 Procedure 1: Connectivity test between sites of the same organization

The requested test confirms connectivity between sites of the same organization. When sourced from the Loopback, Blue1 reaches Blue2 and Red1 reaches Red2. Without that source, the ping fails because the CE-to-PE prefixes are not advertised as VPN networks; the Loopbacks represent the customer networks transported by the L3VPN.

```
Blue1#ping 192.168.102.1 source Loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.101.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/85/104 ms

Blue1#ping 192.168.102.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Listing 41: Blue ping sourced from the Loopback and comparison without Loopback source.

```
Red1#ping 192.168.112.1 source Loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.112.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.111.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/69/76 ms

Red1#ping 192.168.112.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.112.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Listing 42: Red ping sourced from the Loopback and comparison without Loopback source.

The cross-tenant isolation between tenants was also validated. Blue1 does not reach Red2, and Red1 does not reach Blue2. The U code indicates an unreachable destination, because each VRF contains only its own routes.

```
Blue1#ping 192.168.112.1 source Loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.112.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.101.1
UUUUU
Success rate is 0 percent (0/5)
```

Listing 43: Isolation test from Blue1 to Red2.

```
Red1#ping 192.168.102.1 source Loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.111.1
UUUUU
Success rate is 0 percent (0/5)
```

Listing 44: Isolation test from Red1 to Blue2.

3.3 Procedure 2: Routing tables on PE1

The global table on PE1 contains only the provider core: 222.222.1.0/24, the local Loopback 1.1.1.1/32, and OSPF routes to 222.222.2.0/24 and 2.2.2.2/32. The customer networks do not appear in this table.

```
PE1#show ip route
Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
O       222.222.2.0/24 [110/20] via 222.222.1.2, FastEthernet0/0
    2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/21] via 222.222.1.2, FastEthernet0/0
C       222.222.1.0/24 is directly connected, FastEthernet0/0
```

Listing 45: PE1 global table.

The blue VRF contains only blue prefixes: local access, a local static route, and a BGP VPNv4 route to the remote site. The red VRF follows the same logic with red prefixes.

```
PE1#show ip route vrf blue
Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
C       10.10.1.0 is directly connected, FastEthernet1/0
S       192.168.101.0/24 [1/0] via 10.10.1.100
B       192.168.102.0/24 [200/0] via 2.2.2.2

PE1#show ip route vrf red
Gateway of last resort is not set

    11.0.0.0/24 is subnetted, 1 subnets
C       11.11.1.0 is directly connected, FastEthernet1/1
S       192.168.111.0/24 [1/0] via 11.11.1.100
B       192.168.112.0/24 [200/0] via 2.2.2.2
```

Listing 46: Routing tables of the VRFs on PE1.

The isolation results from VRF separation and the Route Target policy: RT 1:1 is imported by the blue VRF; RT 2:2 by the red VRF.

3.4 Procedure 3: ICMP/MPLS capture in the provider network

During the intra-tenant pings, traffic was captured on the PE1-P link. The Echo Requests have two labels: the outer label 16, used as the transport label up to the egress PE, and the inner label 18 for blue or 19 for red, used as the VPN label.

frame.number	ip.src	ip.dst	mpls.label	mpls.bottom	icmp.type
4	192.168.101.1	192.168.102.1	16,18	0,1	8
5	192.168.102.1	192.168.101.1	18	1	0
6	192.168.101.1	192.168.102.1	16,18	0,1	8
7	192.168.102.1	192.168.101.1	18	1	0
34	192.168.111.1	192.168.112.1	16,19	0,1	8
35	192.168.112.1	192.168.111.1	19	1	0
36	192.168.111.1	192.168.112.1	16,19	0,1	8
37	192.168.112.1	192.168.111.1	19	1	0

Listing 47: Wireshark excerpt of the ICMP/MPLS packets captured on the PE1-P link.

The `traceroute` confirms the capture: blue uses 16/18 and red uses 16/19 on the provider hop.

```
Blue1#traceroute 192.168.102.1 source Loopback0 probe 1 timeout 2
Tracing the route to 192.168.102.1
 1 10.10.1.1 28 msec
 2 222.222.1.2 [MPLS: Labels 16/18 Exp 0] 80 msec
 3 10.10.2.1 [MPLS: Label 18 Exp 0] 136 msec
 4 10.10.2.100 80 msec
```

Listing 48: Traceroute with MPLS labels for blue.

```
Red1#traceroute 192.168.112.1 source Loopback0 probe 1 timeout 2
Tracing the route to 192.168.112.1
 1 11.11.1.1 24 msec
 2 222.222.1.2 [MPLS: Labels 16/19 Exp 0] 76 msec
 3 11.11.2.1 [MPLS: Label 19 Exp 0] 80 msec
 4 11.11.2.100 68 msec
```

Listing 49: Traceroute with MPLS labels for red.

The MPLS table on P explains the outer label 16: it points to 2.2.2.2/32 and uses `Pop tag`. Thus, penultimate hop popping (PHP) removes the transport label before the egress PE; the VPN label keeps the VRF identification.

```
P#show mpls forwarding-table
Local  Outgoing  Prefix          Bytes tag  Outgoing     Next Hop
tag    tag or VC  or Tunnel Id    switched  interface
16     Pop tag    2.2.2.2/32      3189      Fa0/1        222.222.2.10
17     Pop tag    1.1.1.1/32      1990      Fa0/0        222.222.1.1
```

Listing 50: MPLS forwarding table on router P.

On PE1, labels 18 and 19 identify the local blue and red networks. For remote routes, PE1 uses the VPN label advertised by PE2.

```
PE1#show ip bgp vpnv4 all labels
Network          Next Hop        In label/Out label

Route Distinguisher: 1:1 (blue)
 192.168.101.0    10.10.1.100     18/nolabel
 192.168.102.0    2.2.2.2         nolabel/18
```

```

Route Distinguisher: 2:2 (red)
  192.168.111.0    11.11.1.100    19/nolabel
  192.168.112.0    2.2.2.2        nolabel/19

PE1#show mpls forwarding-table
Local      Outgoing  Prefix          Outgoing  Next Hop
Label      Label     or Tunnel Id    interface
17         Pop Label 222.222.2.0/24  Fa0/0     222.222.1.2
18         No Label 192.168.101.0/24[V] Fa1/0     10.10.1.100
19         No Label 192.168.111.0/24[V] Fa1/1     11.11.1.100

```

Listing 51: VPNv4 labels and MPLS forwarding on PE1.

The inner VPN label, together with the VRF on the egress PE, makes it possible to separate customers. The RD makes the prefixes unique in MP-BGP, the RT decides the import, and the outer label identifies only the transport LSP, which may change in the provider network.

3.5 Procedure 4: BGP Open and BGP Update capture

To observe the BGP session, the PE1-PE2 neighbor relationship was restarted and the PE1-P link was captured. The BGP Open messages show AS 100, identifiers 1.1.1.1/2.2.2.2, and VPNv4 capabilities.

```
Frame 9: PE1 -> PE2
Border Gateway Protocol - OPEN Message
  Type: OPEN Message (1)
  Version: 4
  My AS: 100
  Hold Time: 180
  BGP Identifier: 1.1.1.1
  Capability: Multiprotocol extensions capability
    AFI: IPv4 (1)
    SAFI: Labeled VPN Unicast (128)
  Capability: Multiprotocol extensions capability
    AFI: IPv4 (1)
    SAFI: Unicast (1)
  Capability: Route refresh capability
  Capability: Support for 4-octet AS number capability
```

Listing 52: Wireshark excerpt of the BGP Open message from PE1 to PE2.

```
Frame 11: PE2 -> PE1
Border Gateway Protocol - OPEN Message
  Type: OPEN Message (1)
  Version: 4
  My AS: 100
  Hold Time: 180
  BGP Identifier: 2.2.2.2
  Capability: Multiprotocol extensions capability
    AFI: IPv4 (1)
    SAFI: Labeled VPN Unicast (128)
  Capability: Multiprotocol extensions capability
    AFI: IPv4 (1)
    SAFI: Unicast (1)
  Capability: Route refresh capability
  Capability: Support for 4-octet AS number capability
```

Listing 53: Wireshark excerpt of the BGP Open message from PE2 to PE1.

The essential capability is Multiprotocol Extension with AFI IPv4 and SAFI Labeled VPN Unicast 128. Without it, the peers do not advertise VPNv4 routes with RD and MPLS label.

In the BGP Update messages, Wireshark shows Extended Communities with Route Target. It also shows MP_REACH_NLRI, Route Distinguisher, and VPN label.

```
Frame 15: PE1 -> PE2
Border Gateway Protocol - UPDATE Message
  Carried extended communities:
    Route Target: 1:1
  Path Attribute - MP_REACH_NLRI
    AFI: IPv4 (1)
    SAFI: Labeled VPN Unicast (128)
    Next hop: RD=0:0 IPv4=1.1.1.1
    Label Stack: 18 (bottom)
    Route Distinguisher: 1:1
    MP Reach NLRI IPv4 prefix: 192.168.101.0

Border Gateway Protocol - UPDATE Message
```

```

Carried extended communities:
  Route Target: 2:2
Path Attribute - MP_REACH_NLRI
  AFI: IPv4 (1)
  SAFI: Labeled VPN Unicast (128)
  Next hop: RD=0:0 IPv4=1.1.1.1
  Label Stack: 19 (bottom)
  Route Distinguisher: 2:2
  MP Reach NLRI IPv4 prefix: 192.168.111.0

```

Listing 54: Wireshark excerpt of the VPNv4 BGP Update messages from PE1 to PE2.

```

Frame 18: PE2 -> PE1
Border Gateway Protocol - UPDATE Message
  Carried extended communities:
    Route Target: 1:1
  Path Attribute - MP_REACH_NLRI
    AFI: IPv4 (1)
    SAFI: Labeled VPN Unicast (128)
    Next hop: RD=0:0 IPv4=2.2.2.2
    Label Stack: 18 (bottom)
    Route Distinguisher: 1:1
    MP Reach NLRI IPv4 prefix: 192.168.102.0

Border Gateway Protocol - UPDATE Message
  Carried extended communities:
    Route Target: 2:2
  Path Attribute - MP_REACH_NLRI
    AFI: IPv4 (1)
    SAFI: Labeled VPN Unicast (128)
    Next hop: RD=0:0 IPv4=2.2.2.2
    Label Stack: 19 (bottom)
    Route Distinguisher: 2:2
    MP Reach NLRI IPv4 prefix: 192.168.112.0

```

Listing 55: Wireshark excerpt of the VPNv4 BGP Update messages from PE2 to PE1.

The BGP VPNv4 table confirms the capture: PE1 received the networks 192.168.102.0/24 and 192.168.112.0/24 from PE2, separated by RD and RT.

```

PE1#show ip bgp vpnv4 all
BGP table version is 7, local router ID is 1.1.1.1

Route Distinguisher: 1:1 (default for vrf blue)
*> 192.168.101.0    10.10.1.100      0          32768 ?
*>i192.168.102.0   2.2.2.2          0    100      0 ?

Route Distinguisher: 2:2 (default for vrf red)
*> 192.168.111.0    11.11.1.100      0          32768 ?
*>i192.168.112.0   2.2.2.2          0    100      0 ?

```

Listing 56: BGP VPNv4 table on PE1 after the MP-BGP session is established.

A VPNv4 route combines RD, IPv4 prefix, Route Target, and MPLS label. Normal BGP IPv4 unicast is not enough because it does not carry the RD, VPN label, or the Route Target import/export policy.

3.6 Conclusion

The BGP/MPLS L3VPN became operational. The underlay uses OSPF and LDP/MPLS; the overlay is signaled by MP-BGP VPNv4. The VRFs isolate the routing tables, the Route Targets control import/export, the inner label identifies the VRF on the egress PE, and the outer label transports the packet through the core. The BGP captures confirm the use of Multiprotocol Extension, MP_REACH_NLRI, Route Target, Route Distinguisher, and VPN label.